

MARWADI UNIVERSITY

DEPARTMENT OF AI, ML

DAA (01AI0506) - Lab Practical

Practical 03: Implementation of Max-Heap Sort Algorithm

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Aim

Implementation of Max-Heap Sort Algorithm as specified in the DAA Lab Manual.

Program

```
#include <iostream>
using namespace std;

void heapify(int a[], int n, int i) {
    int largest = i;
    int left = 2 * i + 1, right = 2 * i + 2;
    if (left < n && a[left] > a[largest]) largest = left;
    if (right < n && a[right] > a[largest]) largest = right;
    if (largest != i) {
        swap(a[i], a[largest]);
        heapify(a, n, largest);
    }
}

void heapSort(int a[], int n) {
    for (int i = n / 2 - 1; i >= 0; i--) heapify(a, n, i);
    for (int i = n - 1; i > 0; i--) {
        swap(a[0], a[i]);
        heapify(a, i, 0);
    }
}

int main() {
    int a[] = {12, 11, 13, 5, 6, 7};
    int n = sizeof(a) / sizeof(a[0]);
    heapSort(a, n);
    for (int x : a) cout << x << " ";
}
```

Output

5 6 7 11 12 13

Time and Space Analysis

Build Heap: $O(n)$. Repeated extraction: $O(n \log n)$. Total Time: $O(n \log n)$. Best/Average/Worst: $\Theta(n \log n)$.
Auxiliary Space: $O(\log n)$ with recursive heapify.

Result

The practical was implemented successfully and the expected output was obtained.